

1 Atomic structure

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Atomic number counts protons; nucleon number counts protons and neutrons. Isotopes have the same proton number but different neutron number. Electron configurations fill orbitals according to energy, Pauli exclusion and Hund's rule. Successive ionisation energy jumps reveal an inner shell after removal of valence electrons.

Worked example: follow the reasoning

Magnesium has $Z = 12$ and electron configuration $1s^2 2s^2 2p^6 3s^2$; Mg^{2+} has $1s^2 2s^2 2p^6$.

Quick check · multiple choice

A What is the electron configuration of Mg^{2+} ?

[1]

A $1s^2 2s^2 2p^6$

B $1s^2 2s^2 2p^6 3s^2$

C $1s^2 2s^2 2p^5$

D $1s^2 2s^2 2p^6 3s^1$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 State electron configuration of Al^{3+} ($Z = 13$) and explain why its third ionisation energy is lower than its fourth.

[3]

Answers and explanation

Quick check: A. $1s^22s^22p^6$

Question 1 [3 marks]

Al^{3+} : $1s^22s^22p^6$ [1]. First three electrons include valence $3s^23p^1$ [1]; fourth is from inner $n = 2$ shell and is held more strongly [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.